Finance Research Letters 67 (2024) 105837



Contents lists available at ScienceDirect

Finance Research Letters

journal homepage: www.elsevier.com/locate/frl





Unveiling tone manipulation in MD&A: Evidence from ChatGPT experiments

Piaopeng Song [a], Hanglin Lu [a,b,*], Yongjie Zhang [a,b]

[a] College of Management and Economics, Tianjin University, Tianjin 300072, China
[b] Complex Management Systems Laboratory, Tianjin University, Tianjin 300072, China

ARTICLE INFO

Keywords:
ChatGPT
Syntactic Complexity
Tone Manipulation
MD&A

ABSTRACT

This research uses ChatGPT-3.5 and ChatGPT-4 to investigate tone manipulation in the management discussion and analysis of annual reports of Chinese-listed companies. We find that the quantification of emotional content in text using financial BERT model and dictionary approach is inconsistent due to two kinds of manipulation: "Expression Manipulation" and "Word Manipulation". Based on ChatGPT we verify the manipulative behavior of complex expressions and sentiment word substitutions. Our research suggests that using ChatGPT with appropriate cue words can help alleviate tone manipulation in financial texts, with GPT-4 having a better effect.

1. Introduction

This research paper validates the presence of "Tone manipulation" in management discussion and analysis(MD&A). It also conducted experiments using the ChatGPT construction strategies and concluded that "Expression Manipulation" is a more significant factor in emotional misinterpretation than "Word Manipulation". Additionally, the study highlights the potential application of ChatGPT-3.5 and ChatGPT-4 in reducing tone manipulation in financial texts to some extent.

Big data has great potential in the financial field. Studies have been conducted on the text data in big data. Scholars have conducted research on the emotional tone tendency of financial texts. The statistical analysis of word frequency based on the dictionary is one of the most commonly used methods for measuring text sentiment (Loughran and Mcdonald, 2011), financial text tone studies are often based on this quantified method. However, both listed company management and financial practitioners may engage in tone manipulation when writing reports (Zhang et al., 2022), due to their professional sensitivity and value orientation. (Huang et al., 2014; Lang and Lundholm, 2000). However, management, constrained by regulatory pressure, is still required to disclose risks, making negative messages identifiable at the sentence level. Our analysis of the MD&A also confirms this perspective, with the statistical results in Table 1 further substantiating this claim. It is evident that there is a significant inconsistency in emotion and tone within negative sentences. Linguistic studies suggest that humans convey more information using shorter sentences when communicating information (Piantadosi et al., 2011), and negative words contain more information than positive ones (Garcia et al., 2012). Therefore, the avoidance of negative words when expressing negative messages can be considered a linguistic anomaly, as it deviates from the norm, as shown in Fig. 1. Starting from this linguistic behavior, we aim to explore this anomaly in the context of financial language.

In order to explore the origins of the aforementioned linguistic anomalies, we conducted experiments involving deconstruction and reconstruction of the text. GPT-3.5-Turbo is a language generation model created by OpenAI, which uses the Transformer architecture (Brown et al., 2020). In experiments involving deconstruction and reconstruction of unstructured textual data, ChatGPT has emerged

* Corresponding author.

https://doi.org/10.1016/j.frl.2024.105837
Received 2 March 2024; Received in revised form 31 May 2024; Accepted 9 July 2024
Available online 14 July 2024
1544-6123/© 2024 Published by Elsevier Inc.

Table 1
This table shows the statistics on tone manipulation in MD&A. Sentence sentiment is recognized based on BERT. Tone is determined by the frequency statistics of words in the financial sentiment dictionary. SC represents syntactic complexity, where Mean stands for the mean and Std represents the standard deviation. Percentage indicates the proportion of this type of sentiment sentence. Consistency reflects whether the sentiment of a sentence aligns with the tone. "Inconsistent" means the opposite of sentence sentiment and intonation judgment result, "Consistent" means the same of sentence sentiment and intonation judgment result.

Sentence Sentiment	Tone	Count	SC Mean	SC Std	Percentage	Consistency
Positive	Tone >0	1,665,812	17.246	8.954	78.9 %	Consistent
	Tone <0	28,228	17.700	10.400	1.4 %	Inconsistent
Negative	Tone <0	245,750	19.741	11.507	11.7 %	Consistent
	Tone >0	169,623	20.811	12.219	8.0 %	Inconsistent

as a widely utilized tool which has demonstrated excellent capabilities in various domains and tasks (Ye et al., 2023) . The capabilities and testability of GPT have been demonstrated in the field of finance (Leippold, 2023). The study aims to utilize ChatGPT to examine whether anomalous alterations in MD&A language stem from deliberate expression manipulation or merely the substitution of emotive vocabulary.

2. Experimental method

2.1. Measurement and indicator calculation

2.1.1. Sentiment recognition

The BERT deep learning model is used in this paper to determine the emotional tone of sentences. Leippold (2023) has confirmed the reliability of such methods, which are highly effective in preventing tone manipulation. In the word manipulation group experiment, the FinBERT model (Araci, 2019; Liu et al., 2020) used in this paper achieved an excellent F1 score of 0.951. Thus, the BERT class model is also used in this study. Since this paper focuses on the MD&A data of Chinese listed companies, the "Finance Sentiment ZH"[1] BERT sentiment model for Chinese financial texts is selected. This model is open source and was developed by BardsAI Company on Huggingface. It is a Bert-Base-Chinese model that analyzes the sentiment of Chinese financial news. The model was trained on the Chinese translation version of Malo et al. (2014)'s Financial Phrase Bank for 10 rounds and has a good performance with an F1 score of 0.966. This paper utilizes this model to determine the emotional tone of financial statements.

2.1.2. Tone

Financial sentiment dictionary methods are commonly used in sentiment analysis of financial texts, such as the LM Dictionary (Loughran and Mcdonald, 2011). These methods rely on dictionaries to measure the sentiment of statements by counting the presence or absence of specific words or phrases. In this paper's experiment, our focus is on the MD&A of listed companies in China because the supervision of MD&A in the annual reports of listed companies in China needs improvement. Moreover, the diversity of the Chinese language makes it more prone to manipulation. Therefore, we chose a dictionary developed by Yao et al. (2021), which is more suitable for the sentiment dictionary in the Chinese financial field and widely used in the field of Chinese financial texts. Tone, reference formula for calculating sentence intonation index based on keyword method (1) (Huang et al., 2014; Loughran and Mcdonald, 2011).

$$Tone = \frac{n_{positive} - n_{negative}}{n_{positive} + n_{negative}} \tag{1}$$

2.1.3. Readability

For the purpose of quantifying the readability of text, we use two methods.

Syntax Complexity: Miller (2010) suggests that SC(Syntactic Complexity Index), which represents the average number of words in a sentence, can indicate the readability of sentences. The higher the value of SC, the more complex the syntax and the lower the readability.

Perplexity: Perplexity serves as an essential measure to assess the efficacy of language models by computing the probability of sentences. A logical sentence will yield a high probability when assessed by a language model (Shin et al., 2020). This study employs the BERT mask probability calculation technique to evaluate Perplexity.

BERT is a Masked Language Model (MLM) that calculates the sum of probabilities of all masked words based on the context of the whole sentence processed by the mask.

$$\log p_\theta(\overline{\mathbf{x}}|\widehat{\mathbf{x}}) \approx \sum_{t=1}^{T} m_t \log p_\theta(x_t|\widehat{\mathbf{x}}) = \sum_{t=1}^{T} m_t \log \frac{\exp\left(H_\theta(\widehat{\mathbf{x}})_t^\top e(x_t)\right)}{\sum_{x'} \exp\left(H_\theta(\widehat{\mathbf{x}})_t^\top e(x')\right)} \tag{2}$$

Then calculate the Perplexity as (3):

[1] https://huggingface.co/bardsai/finance-sentiment-zh-base

$$Perplexity = \frac{1}{N}\sum_{s=1}^{N}\log P_s(P_w) \tag{3}$$

2.1.4. Inconsistent of Tone and sentiment

We define tone manipulation as the incongruity between the linguistic tone of a sentence and its sentiment disposition. To investigate if "Tone Manipulation" exists in MD&A, and to conduct quasi-natural experiments to identify the source, we employed the following NLP methods: financial sentiment dictionary method, BERT (Devlin et al., 2019) deep learning model (for sentence emotional assessment of financial statements), and ChatGPT.

2.2. GPT-3.5 experimental strategy and index

GPT is a large-scale language model that has been trained on a significant amount of textual data. It comprehends the meaning of sentences through a self-attention structure and can fully understand the context of sentences and provide responses using autoregressive modules. As a new analytical tool, ChatGPT can assist in financial research and has significant advantages in data identification and analysis (Dowling and Lucey, 2023). ChatGPT is utilized in a wide range of financial sectors. The technology of ChatGPT can reduce fraudulent activities by analyzing user interactions to minimize the occurrence of fraudulent activities and loss of funds (George et al., A.S. 2023). Additionally, ChatGPT's big data processing capability can analyze the comprehensive profile of borrowers and helps financial institutions identify risks (Wei et al., 2023). ChatGPT can even make personalized investment recommendations based on an individual's financial goals and risk preferences (Oehler and Horn, 2024). Companies can also utilize ChatGPT to improve customer experience and increase consumer satisfaction (Korzynski et al., 2023). Our experiment aimed at deconstruction and reconstruction of unstructured textual data, the study drew inspiration from Leippold (2023) and employed ChatGPT for exploration.

The Prompt strategy was developed with several adjustments based on feedback to ensure it was the most effective strategy. In other words, the experimental results of GPT are validated through manual inspection. Initially, we experimented with a small sample of 100 sentences and evaluated the intonation correction. We then shared the results with 5 graduate students who specialize in corporate finance to assess areas for improvement. Based on their evaluation, we further refined and optimized the prompt words to ensure the construction of prompt words was rigorous (Yong et al., 2023; Polak and Morgan 2024). The strategies we have devised is as follows:

Strategy 1: Prompt of Simplified Expression

Strategy 1 we set the Prompt word as: "Play a professional and cautious financial analyst. I will give you the contents of the annual report, please express it more understandably". We define this Strategy as ReExpr.

Strategy 2: Prompt of Generalization of Tone words

Strategy 2 we set the Prompt word as: "Play a professional and cautious financial analyst. I will give you the contents of the annual report. Please rewrite the contents by replacing the emotional words with more common emotional expressions". We define this Strategy as ReWord.

Both strategies to access ChatGPT-3.5 frequently involve calling OpenAI's API. We set the temperature parameter at 0.5 to control the diversity and randomness of generated text.

2.3. Data

We used the Management Discussion and Analysis (MD&A) data from the annual reports of Chinese A-share listed companies. The data was collected from 2010 to 2022, and the source was CSMAR. To avoid any unusual effects in the sample, we applied some filters. Firstly, we excluded ST stocks. Secondly, we removed companies that have been listed for less than a year and have been suspended or delisted. Thirdly, we eliminated companies listed on the Beijing Stock Exchange. We also excluded listed companies in the financial sector. These sample screening practices are commonly used in corporate finance research to avoid sample bias (Huang et al., 2014).

We have made pauses in the MD&A from a chosen sample (Extract each complete sentence), using periods, question marks, exclamation marks, and semicolons as our standard. We have also taken into consideration the data cleaning process and any unusual statements. After cleaning, we have selected experimental samples consisting of retained statements with a length of less than 100 characters.

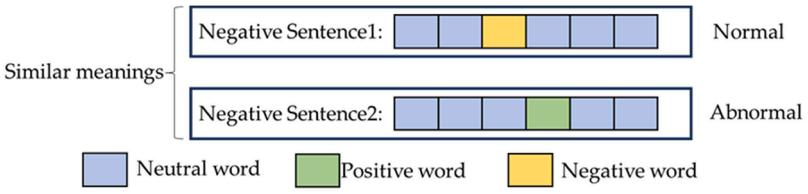


Fig. 1. Sentence1 and Sentence2 contain the same negative message but differ in word structure. For normal language behavior, Sentence1′s expression habit should be followed. If the language wants to euphemize or dilute the language information density, Sentence2′s way will be followed.

3. Experimental result

3.1. Sample data statistics

It has been observed that the proportion of inconsistencies in negative sentences is higher than that in positive sentences, regardless of the type of same sentiment or all sentiment. According to Table 1, although Positive Sentence accounts for a larger proportion of all emotional sentences, Inconsistent is more common in Negative Sentence, with Positive(Tone<0) and Negative(Tone>0) accounting for 1.4 % and 8.0 % respectively. This suggests that a greater amount of tone manipulation occurs even in statements with fewer negative emotions, which is a widespread phenomenon. In addition, the proportion of inconsistent sentences in positive sentences is only 1.7 % of the same emotion(1.7 % may be due to data noise), whereas in negative sentences, it is 40.6 %. This indicates that there are fewer inconsistencies in positive sentences and more intonation manipulation in negative ones. Therefore, it can be concluded that companies use MD&A to manipulate emotions by avoiding the use of common negative words in their annual reports. Furthermore, the large proportion of manipulated negative sentimental statements suggests that most management deliberately avoids using common negative words when writing negative news in the annual report, to reduce the probability of investors discovering the true state of the company.

After analysing sentence in MD&A, we have found that statement groups with inconsistent statements have a higher SC value than groups with consistent statements. This phenomenon is more evident in groups with negative sentences. According to the statistics of Table 1, inconsistent versus consistent have higher SC values for the same type group of sentiment in MD&A statements with non-neutral sentiment. We conducted a T-test to further prove the significance of this difference, and the result is displayed in Fig. 2. In MD&A, both positive and negative sentences have significantly higher SC for inconsistent groups. The |T| value of consistent and inconsistent in negative sentences is 28.7, which is much higher than the value of |T| 8.41 in positive sentences. This indicates that there are more significant differences in negative sentences. Therefore, it can be concluded that MD&A changes the expression of negative sentences in the annual report to enhance the complexity of syntax, which helps to cover negative sentiment and achieve intonation manipulation.

3.2. ChatGPT-3.5 experiment

In the ChatGPT-3.5-based experiment, we conducted a more comprehensive study to verify the preliminary reasons for manipulation mentioned earlier. Due to the high cost of the ChatGPT-3.5 experiment, we randomly selected 10,000 samples of Negative Sentences with an inconsistent tone (Tone=1) for verification, which accounted for around 15.2 % of the entire dataset. We used the ChatGPT-3.5-based generative artificial intelligence to determine whether the negative tone manipulation in the MD&A section of the annual report was based on "Expression Manipulation" or "Word Manipulation". The results of the experiment are presented in Fig. 3.

Based on the text statistics before and after ChatGPT-3.5-Turbo experiment, we found that both Prompt strategies can help reduce emotional misjudgement. The ReExpr-based Prompt reduces the Tone to 0.768, and the Reword-based prompt reduces the Tone to 0.816. A paired sample T-test between the two groups and the original data was significant ($P < 0.01$). We also compared syntactic complexity before and after ChatGPT-3.5 Turbo processing, and found that the ReExpr Prompt strategy significantly reduced syntactic complexity ($t = 75.3$, $P < 0.01$, reduced from 23.38 to 12.20), Perplexity (The average fell to 3.03 from 7.74). By comparing the results before and after ReExpr strategy treatment, this further proves that the company's management changes the expression ways of negative emotion statements to reduce the readability and enhance the syntax complexity to cover up negative emotions. Although the ReWord Prompt strategies can also significantly reduce syntactic complexity ($t = 57.0$, $P < 0.01$, reduced from 23.38 to 14.48),

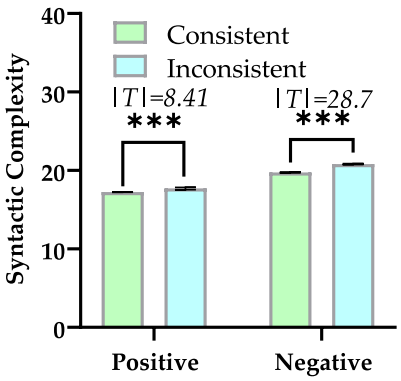


Fig. 2. This figure presents the syntactic complexity of tone-consistent and tone-inconsistent sentences, categorized by positive and negative sentiment sentences. The error bars represent 95 % confidence intervals. The *t*-test was conducted to examine the differences in syntactic complexity between the consistent and inconsistent groups in both positive and negative sentiment sentences. The results revealed significant variations. Specifically, in positive sentiment sentences, the inconsistent group exhibited a significantly higher syntactic complexity of 0.454 compared to the consistent group. Similarly, in negative sentences, the inconsistent group showed a significantly higher syntactic complexity of 1.07 compared to the consistent group.

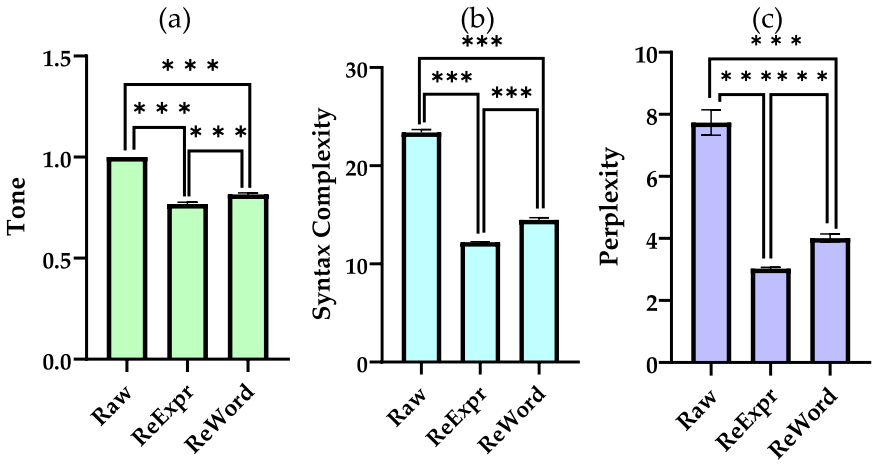


Fig. 3. This figure shows the text metric statistics before and after ChatGPT-3.5 processing. Sample selection is comprised of 10,000 randomly chosen sentences from Inconsistent negative sentences (approximately 15.2 % of the total). Raw represents the original sample sentences. ReExpr refers to text processing using ChatGPT-3.5-Turbo combined with the first prompt strategy, which simplifies expression. ReWord, on the other hand, involves text processing using ChatGPT-3.5-Turbo combined with the second prompt strategy, which generalizes tone words. Subgraph (a) shows the differences in Tone for the three groups, and paired *t*-tests indicate significant differences in the means between each pair of groups. Subgraph (b) illustrates the syntactic complexity of the three groups, and paired *t*-tests reveal significant differences in the means between each pair of groups. Notably, ReExpr has lower syntactic complexity. Subgraph (c) analysing the perplexity degree of the three groups of sentences, statistical results indicate that the ReExpr strategy has a lower perplexity degree and it can significantly improve the readability of manipulative statements with negative messages.

Perplexity (The average fell to 4.01 from 7.74), it is not as effective as the ReExpr Prompt strategy.

3.3. Robustness discussion

1. Dictionary replacement

Formula (1) represents the definition of "Tone" based on the lexicographic approach proposed by Yao et al. (2021), aiming to demonstrate the robustness of Tone and the robustness of the experimental conclusions in this article. In this study, an additional method of lexical measurement was introduced. We employed an alternative financial lexicon to measure the tone before and after the experiment, specifically using an updated, more comprehensive Chinese financial sentiment lexicon developed by Jiang et al. (2021) based on the Loughran and MacDonald (2011) lexicon, which was expanded through manual curation and the utilization of the word2vec algorithm. The results presented in Table 2 indicate that the manipulation of expression in statements is effectively reduced by ReExpr and ReWord, with ReExpr demonstrating a more significant efficacy.

2. Experiment with GPT4

Compared to GPT3.5, GPT4 has a better understanding and analysis ability of financial knowledge. However, due to the high cost of the experiment, we only randomly selected 1000 negative emotion statements with manipulated intonation from the samples in the GPT3.5 experiment. The results are reported in Table 3, and statistics show that using the same strategy, GPT4 has a better ability to reduce intonation manipulation and improve readability than GPT3.5.

4. Conclusion

The paper's empirical and experimental results suggest that management manipulates both expression and wording in management discussion and analyses. One side, companies often try to hide negative emotions by using more complex language. The other side, they also tend to avoid using negative words and instead use more positive language to emphasize their efforts to solve problems and improve the situation. It should be emphasized that the most management achieves its goals through covert expression manipulation, which can easily go unnoticed. When management receives news of negative emotions, they tend to alter the way it is expressed. It's important to note that even if negative news is delivered in a positive tone, it doesn't necessarily mean that the underlying issue has been resolved. Investors and shareholders should approach the company's prospects with caution and keep in mind that positive language from management may not always reflect the reality of the situation. Our study revealed that using ChatGPT-3.5-Turbo appropriately along with the right prompt words can somewhat reduce the verbal and emotional inconsistencies caused by manipulation. This makes ChatGPT-3.5 and ChatGPT-4 more effective in simplifying language expression compared to replacing emotional words. ChatGPT-3.5-Turbo and ChatGPT-4 also help to decrease statement complexity and improve text readability. This experiment can be used as a reference for the financial industry to process text data using ChatGPT.

Table 2
This table presents the statistical analysis of the substitution vocabulary (Jiang et al., 2021) for the word "tone".

	Raw	ReExpr	ReWord
Tone	0.472	0.337	0.405

Table 3
This table shows the ChatGPT-4′s experimental results.

	Raw	ReExpr	ReWord
Tone	1.0	0.663	0.717
AvgLength	23.336	12.220	14.133
PPL(Perplexity)	7.454	2.781	4.329

CRediT authorship contribution statement

Piaopeng Song: Writing – review & editing, Validation, Project administration, Formal analysis, Data curation. **Hanglin Lu:** Writing – review & editing, Writing – original draft, Visualization, Software, Formal analysis, Data curation, Conceptualization. **Yongjie Zhang:** Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Data availability

Data will be made available on request.

Funding

Funded by the National Natural Science Foundation of China, Project Number 72342022.

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